

Okibe (Emmanuel) Abang

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EDUCATION

University of Calgary, Schulich School of Engineering

Calgary, Canada

Bachelor of Science – Software Engineering

June 2026

Relevant Coursework: Data Structures & Algorithms, Database Management, Software Architecture, Operating Systems, Statistics & Machine Learning, Computer Networking, Software Testing, Software Performance Evaluation, Computer Architecture.

CERTIFICATIONS

- AWS Practitioner
- CompTIA Security +

SKILLS

Programming: Python, JavaScript, TypeScript, Java, C++, C, HTML, CSS

Frameworks & Libraries: React, Node.js, Express, Django, Flask, Tailwind CSS, Socket.IO, Scikit-learn, Pandas, NumPy

Databases: MySQL, PostgreSQL, MongoDB, DynamoDB (NoSQL)

Cloud & DevOps: AWS (DynamoDB, Lambda, API Gateway, S3, CloudWatch), Docker, Terraform, CI/CD

Tools: Git, GitHub/GitLab, Jira, Postman, Figma, VS Code

Concepts: REST APIs, OOP, Role-Based Access Control (RBAC), JWT Auth, Real-Time Systems (WebSockets), MVC Architecture, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineering Consultant (Capstone)

Sep 2025 – Apr 2026

Al Nasera Electronics | Calgary, Canada | React.js, Node.js, Express, MySQL, bcrypt, Docker

- Embedded in a 6-person engineering team contracted by Al Nasera Electronics to fully redesign and modernize their internal job management platform, working directly alongside the client's software engineers across a 3-sprint Agile cycle.
- Contributed to backend REST API development covering the complete job lifecycle: job creation, warranty evaluation, technician examination, quotation generation, customer approval, parts handling, repair, and job closure.
- Designed and executed data migration scripts to transfer all records from 2020 onward into a new MySQL schema with 100% accuracy, including data standardization across phone formats, ISO dates, and schema normalization.
- Implemented RBAC enforcing distinct dashboards and access rules for admin, operator, and technician roles, blocking 100% of unauthorized access attempts validated through penetration testing.
- Participated in unit, integration, end-to-end, and UAT testing cycles with the client team; delivered comprehensive deployment documentation enabling the sponsor to independently run and maintain the system.

PROJECTS

DocIntel — AI-Powered Document Q&A Platform

Personal project | React, Node.js/Express, PostgreSQL (pgvector), AWS (S3, ECS, Lambda), Docker, Terraform, GitHub Actions

- Built a full-stack RAG (retrieval-augmented generation) platform enabling users to upload documents and receive grounded, citation-backed answers to natural-language questions.
- Designed a document processing pipeline handling text extraction, chunking, and embedding generation, with vector similarity search implemented via PostgreSQL/pgvector.
- Deployed the application on AWS using Terraform-provisioned infrastructure (ECS Fargate, S3, RDS) and a GitHub Actions CI/CD pipeline for automated build, test, and deploy.
- Implemented streaming LLM responses with inline source citations, reducing hallucinated answers by grounding all outputs in retrieved document context.

PlayPal – Social Activity Matching Platform

Jan 2025 – Apr 2025

Team project (5 members) | React 18, Node.js, Express, MongoDB, Socket.IO, Google Gemini AI, Docker | SENG 513

- Led backend database architecture and real-time messaging infrastructure for a social platform enabling users to post activities, find partners, and chat in real time.
- Designed and implemented MongoDB schemas for users, posts, conversations, messages, and ratings; enforced role-based access control (user/host/admin) validated server-side on every request.

- Built the real-time chat system using Socket.IO with per-conversation rooms, persistent message history, and an AI co-pilot (Gemini) with summarize, decide, and poll modes.
- Developed a friend system, notification pipeline, and a rating system with prompt chips; integrated Google Gemini to generate personalized activity match explanations with client-side caching.
- Coordinated integration between the database layer and frontend teammates, maintaining consistent API contracts and resolving cross-module bugs throughout the sprint cycle.

Emotion Detection – CNN vs. Transfer Learning (ResNet50)

Jan 2025 – Apr 2025

Team project | Python, TensorFlow/Keras, Scikit-learn, NumPy, Pandas, OpenCV | ENSF 519

- Built and trained a Convolutional Neural Network from scratch on the AffectNet dataset for real-time facial emotion detection, targeting a mental-health support use case for virtual therapy sessions.
- Implemented the full preprocessing pipeline: face detection, cropping, resizing, normalization, and augmentation to handle class imbalance across 8 emotion categories.
- Ran systematic training experiments tuning batch size, learning rate, and architecture depth; compared CNN performance against a ResNet50 transfer learning model using accuracy, loss curves, and confusion matrices.
- Identified that transfer learning significantly outperformed the from-scratch CNN on limited data, producing actionable insights on model selection for constrained real-world datasets.

LEADERSHIP

Technical Lead – PlayPal Backend & Real-Time Systems

Jan 2025 – Apr 2025

- Took ownership of the backend architecture and database layer for a 5-person team project, defining data models, API contracts, and real-time messaging infrastructure used across the entire application.
- Coordinated with frontend teammates to maintain consistent integration points and unblocked cross-team issues throughout sprint cycles, ensuring timely delivery of all milestones.
- Presented a deep dive on WebSockets and Socket.IO architecture to the team, including Q&A on scaling strategies and heartbeat mechanisms for production readiness.